



Radiotherapy

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Radiotherapy

- Medical use of ionizing radiation to treat cancer
- Aims to destroy malignant cells
- Minimizes damage to normal tissues
- Used alone or with surgery and chemotherapy



Purpose of Radiotherapy

- Kill cancer cells by damaging DNA
- Control tumor growth
- Reduce recurrence risk
- Improve patient survival

History of Radiotherapy

- Discovery of X-rays in 1895
- First cancer treatments using radiation
- Development of radioactive sources
- Introduction of linear accelerators

Importance of Radiotherapy

- Used in about half of cancer cases
- Improves local tumor control
- Reduces pain and symptoms
- Enhances treatment outcomes



Types of Radiotherapy

- External beam radiotherapy
- Brachytherapy
- Systemic radiotherapy

External Beam Radiotherapy

- Radiation delivered from outside the body
- Most commonly used technique
- Uses linear accelerators
- Suitable for deep and superficial tumors



Brachytherapy

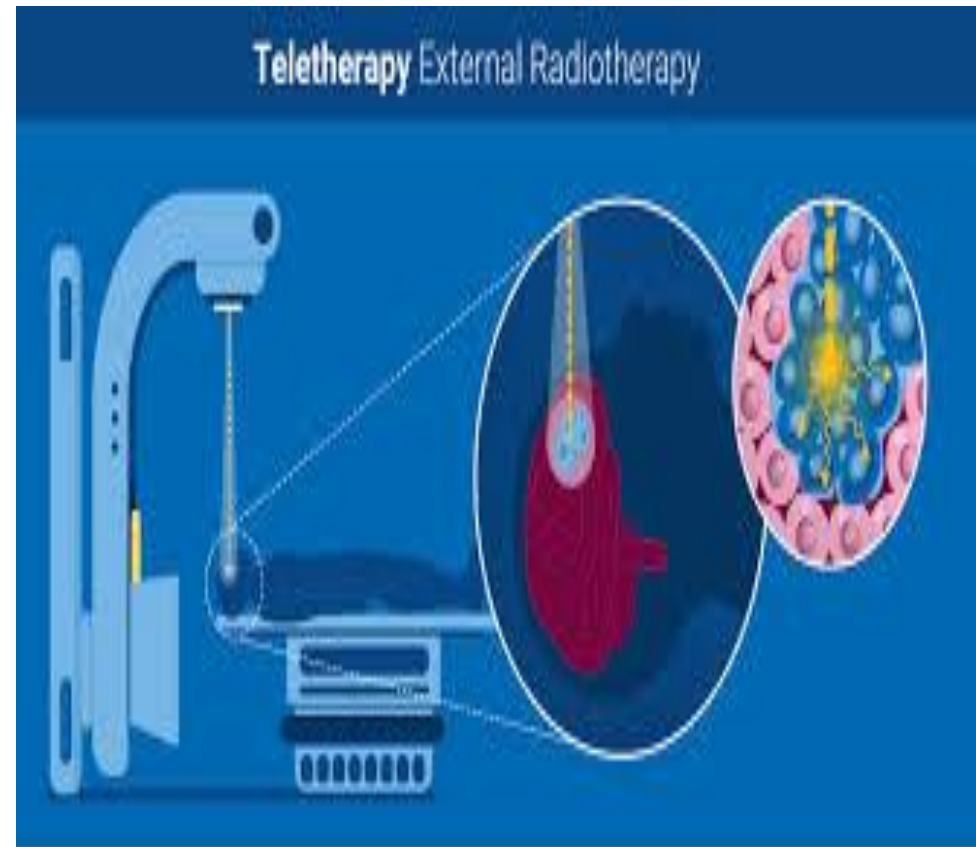
- Radioactive sources placed inside the body
- High dose to tumor
- Low dose to surrounding tissues
- Used in prostate and cervical cancer

Systemic Radiotherapy

- Radioactive drugs given orally or intravenously
- Travel through bloodstream
- Target specific organs or tumors
- Example: Radioiodine for thyroid cancer

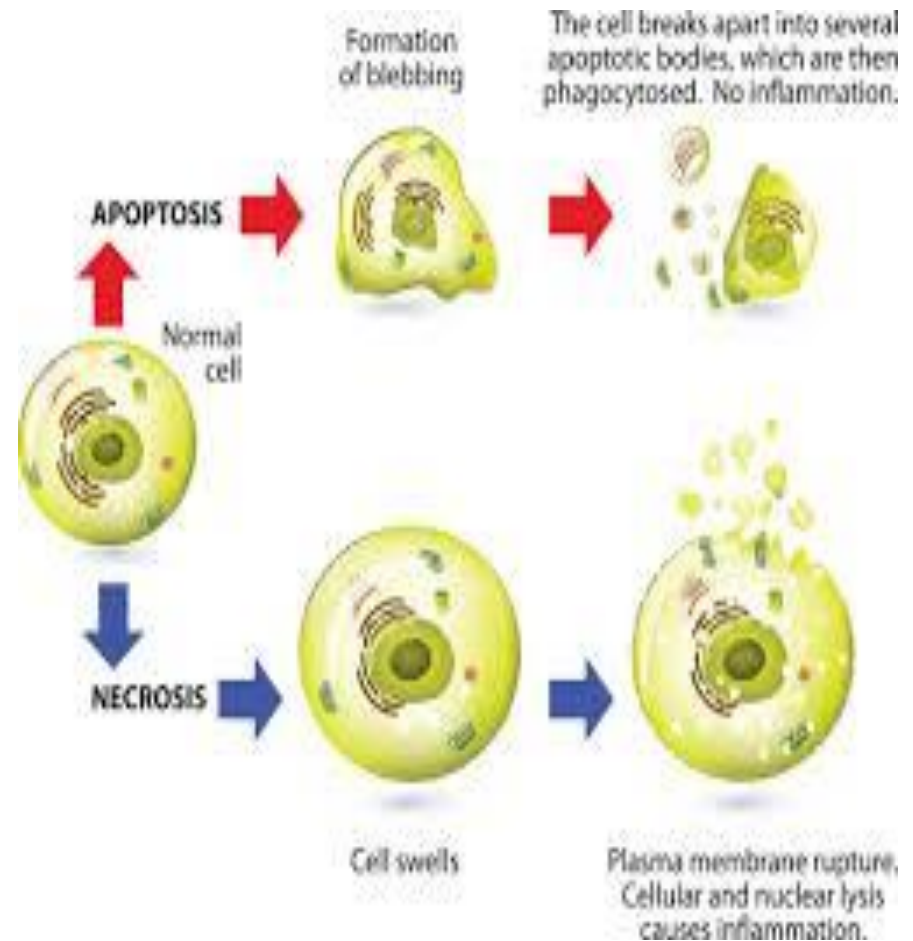
Biological Effect of Radiation

- Damages cellular DNA
- Prevents cell division
- Causes cancer cell death
- Normal cells recover better



Mechanisms of Cell Death

- Apoptosis
- Mitotic catastrophe
- Necrosis
- Loss of reproductive ability



Fractionation

- Total dose divided into sessions
- Allows normal tissue recovery
- Improves tumor control
- Reduces side effects

Benefits of Fractionation

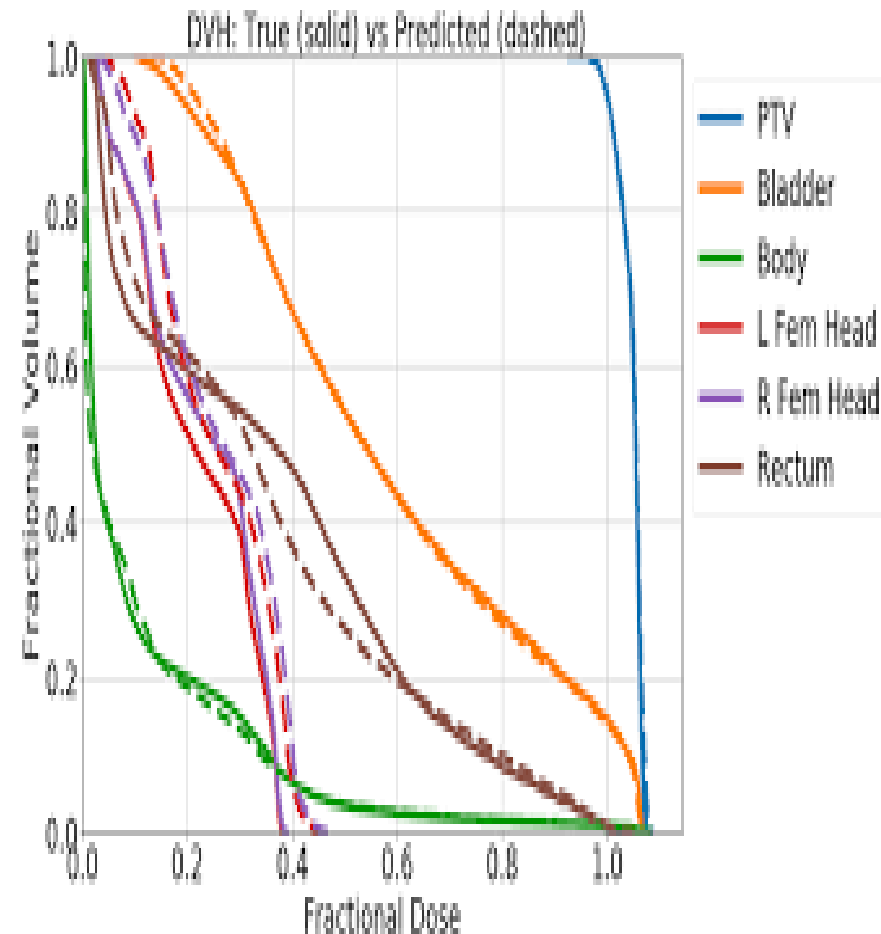
- Better tumor response
- Less damage to healthy tissue
- Improved patient tolerance
- Flexible treatment schedules

Dose-Response Relationship

- Higher dose increases tumor kill
- Limited by tissue tolerance
- Balance between benefit and risk
- Guides treatment planning

Therapeutic Ratio

- Maximize tumor damage
- Minimize normal tissue injury
- Main goal of radiotherapy
- Achieved through precision



Radiotherapy Workflow

- Patient diagnosis
- Imaging and simulation
- Treatment planning
- Dose delivery and follow-up

Introduction to Linear Accelerator (LINAC)

- Machine that accelerates electrons
- Produces X-rays or electron beams
- Used in external beam therapy
- Standard radiotherapy equipment



Role of LINAC in Radiotherapy

- Delivers accurate radiation doses
- Treats deep and superficial tumors
- Supports advanced techniques
- Improves treatment outcomes

Evolution of LINAC Technology

- Early machines in 1960s
- Development of 3D radiotherapy
- Introduction of IMRT and VMAT
- Integration with imaging systems

Comparison with Cobalt-60 Units

- LINAC produces variable energies
- No radioactive source storage
- More precise beam shaping
- Safer and more flexible

Comparison with Proton Therapy

- Proton therapy uses charged particles
- LINAC uses photons and electrons
- Proton therapy is more expensive
- LINAC widely available

Advantages of LINAC

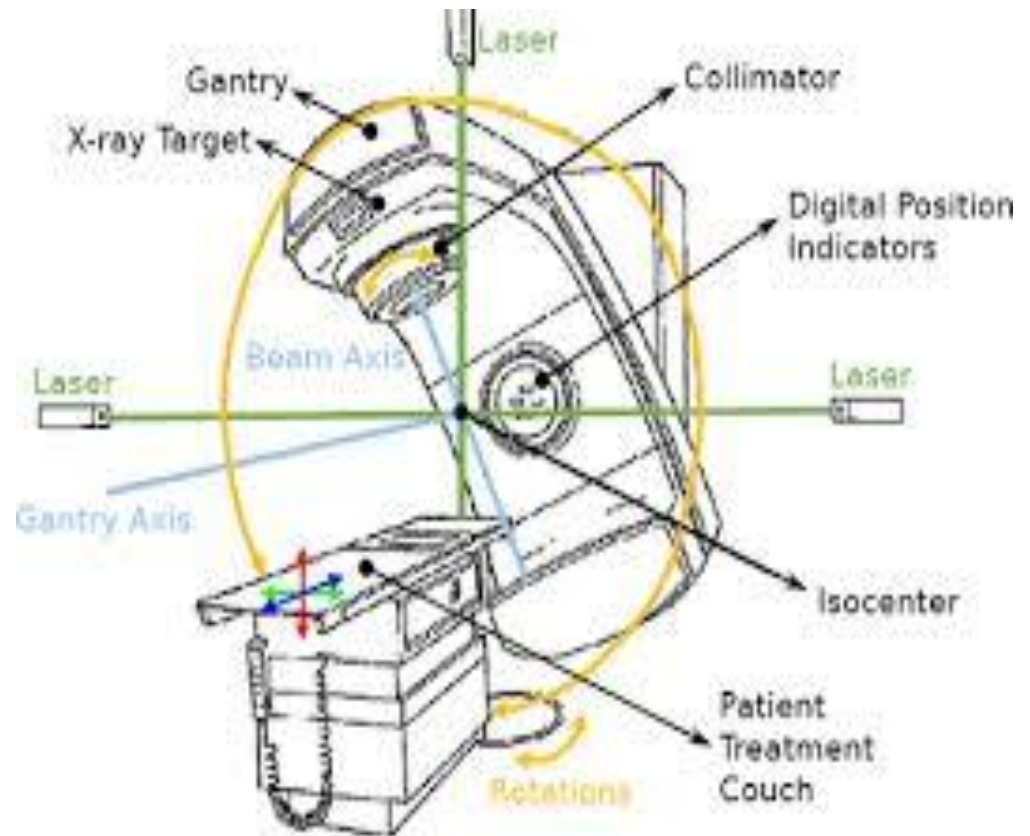
- High precision
- Adjustable beam energy
- Advanced treatment techniques
- No radioactive materials

Basic Physics of LINAC

- Uses electromagnetic waves
- Accelerates electrons
- Produces high-energy radiation
- Converts electrons to X-rays

Main Components of LINAC

- Electron gun
- Waveguide
- Microwave source
- Bending magnet
- Treatment head



Electron Gun

- Produces electrons
- Starts beam generation
- Controls beam current
- Located at LINAC base

Waveguide

- Accelerates electrons
- Uses microwave energy
- Maintains beam direction
- Increases electron energy

Magnetron / Klystron

- Generates microwave power
- Supplies RF energy
- Enables electron acceleration
- Essential LINAC component

Bending Magnet

- Changes beam direction
- Guides electrons to target
- Adjusts energy selection
- Ensures beam accuracy

Gantry

- Rotates around patient
- Allows multiple beam angles
- Improves dose conformity
- Enhances treatment accuracy

Collimators

- Shape radiation beam
- Limit exposure to normal tissue
- Improve targeting
- Adjustable during treatment

Multileaf Collimator (MLC)

- Uses movable metal leaves
- Shapes beam precisely
- Enables IMRT and VMAT
- Improves tumor conformity

Treatment Couch

- Supports patient positioning
- Allows fine adjustments
- Ensures reproducibility
- Improves accuracy

Beam Production Process

- Electrons generated
- Accelerated in waveguide
- Directed to target
- Converted to X-rays

Photon Beams

- Used for deep tumors
- Penetrate body tissues
- Produced by bremsstrahlung
- Main radiotherapy beam

Electron Beams

- Used for superficial tumors
- Limited penetration depth
- Directly delivered electrons
- Suitable for skin lesions

Image-Guided Radiotherapy (IGRT)

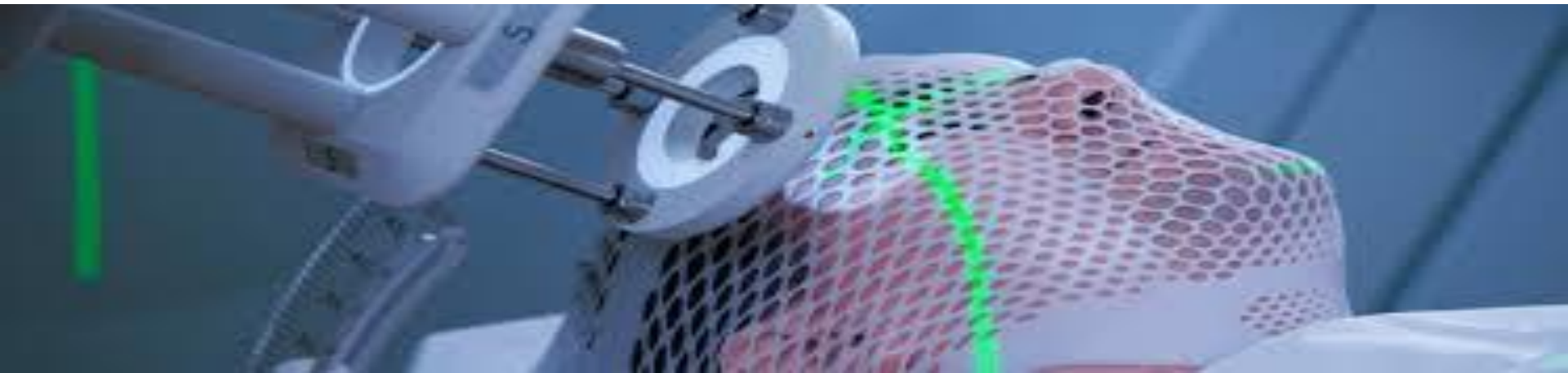
- Uses imaging before treatment
- Improves patient positioning
- Reduces setup errors
- Increases treatment accuracy

Treatment Planning Systems

- Computer software for planning
- Calculates radiation dose
- Optimizes beam arrangement
- Protects organs at risk

Patient Immobilization

- Prevents patient movement
- Improves reproducibility
- Enhances targeting accuracy
- Includes masks and molds

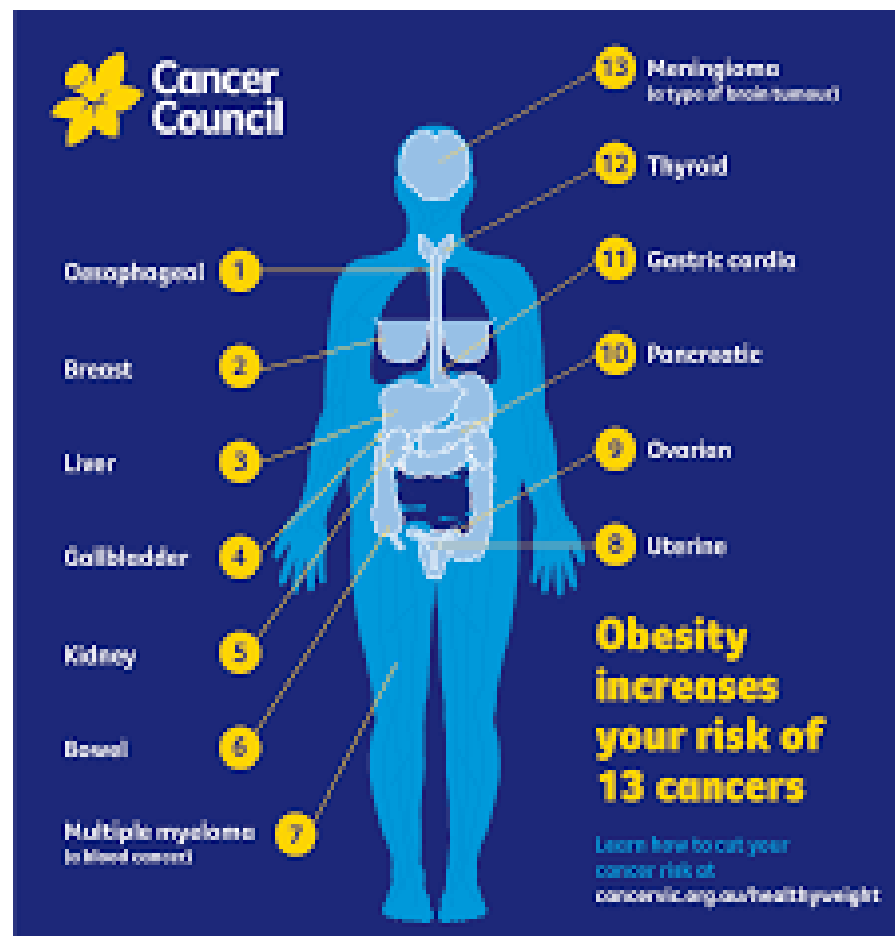


Quality Assurance in Radiotherapy

- Daily machine checks
- Monthly and annual tests
- Ensures dose accuracy
- Improves patient safety

Clinical Applications of LINAC

- Breast cancer
- Lung cancer
- Prostate cancer
- Head and neck cancer



Future of Radiotherapy

- Artificial intelligence integration
- Adaptive radiotherapy
- MR-LINAC systems
- FLASH therapy development